Hotel Revenue & Booking Analytics

A Comprehensive Data Analytics Case Study

2022 – 2023 | 3,000 Bookings | 4 Room Categories

Total Revenue	Avg Daily Rate	Cancellation Rate	Guest Satisfaction
Rs 4.97 Cr	**Rs 7,754**	**23.2%**	**7.43/10**

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Executive Summary

This project analyses hotel booking data across 2022–2023, covering 3,000 reservations spanning four room categories and five booking channels. Leveraging Python (Pandas, Matplotlib, Seaborn, Scikit-learn), the study uncovers revenue drivers, seasonal patterns, cancellation risk factors, and guest satisfaction insights — with a machine-learning model predicting booking cancellations.

Total Revenue	Avg Daily Rate	Avg Stay	Cancel Rate	Satisfaction	Repeat Guests
Rs 4.97 Cr	**Rs 7,754**	**2.76 nights**	**23.2%**	**7.43/10**	**27.1%**

Key Findings

- August is the peak revenue month; December shows a strong holiday spike.
- OTA (Online) is the largest booking channel but carries the highest cancellation rate.
- Suite guests report an average satisfaction of 8.5/10, the highest of all room types.
- Direct bookings yield 0.3 points higher satisfaction vs OTA on average.
- Lead time is the strongest predictor of cancellation risk (Random Forest model).
- RevPAR peaks in July–August and dips in Jan–Feb, aligning with travel seasonality.

1. Revenue Analysis

1.1 Monthly Revenue Trend

Revenue demonstrates a clear seasonal pattern over the two-year study period, with twin peaks each year in July–August (summer leisure travel) and December (festive season). Revenue troughs occur in January–February, suggesting opportunities for targeted promotional packages.

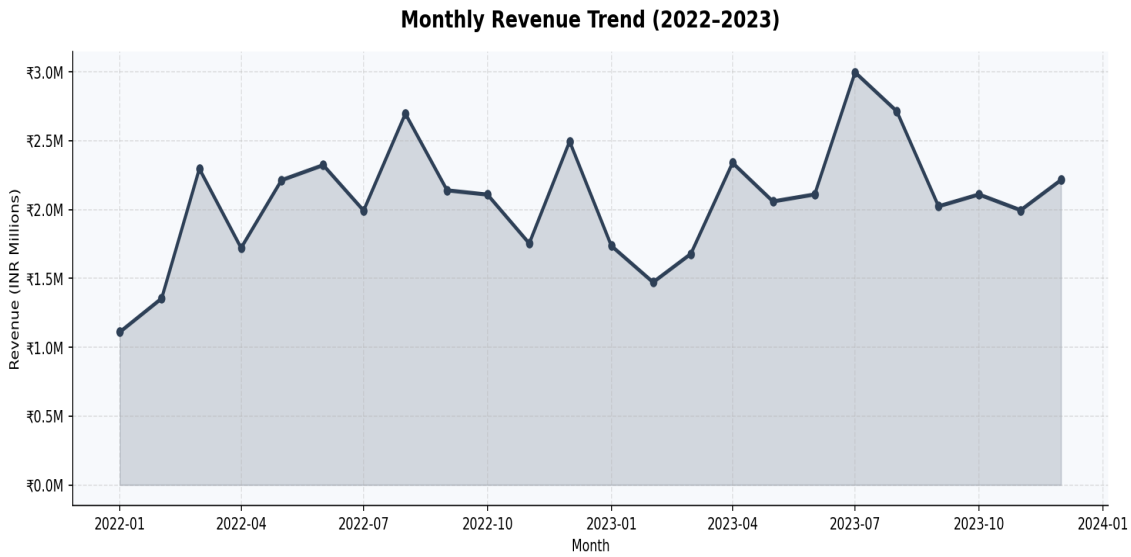

Fig 1.1 — Monthly Revenue Trend (2022–2023)

Insight: Peak month is Aug both years. Consider launching 'Off-Peak Value Packages' in Jan–Feb to smooth revenue.

1.2 Revenue by Room Type

Despite having the lowest ADR, Standard rooms generate the highest absolute revenue due to volume. Suite rooms carry the highest ADR and offer strong upsell potential, contributing disproportionate revenue per booking.

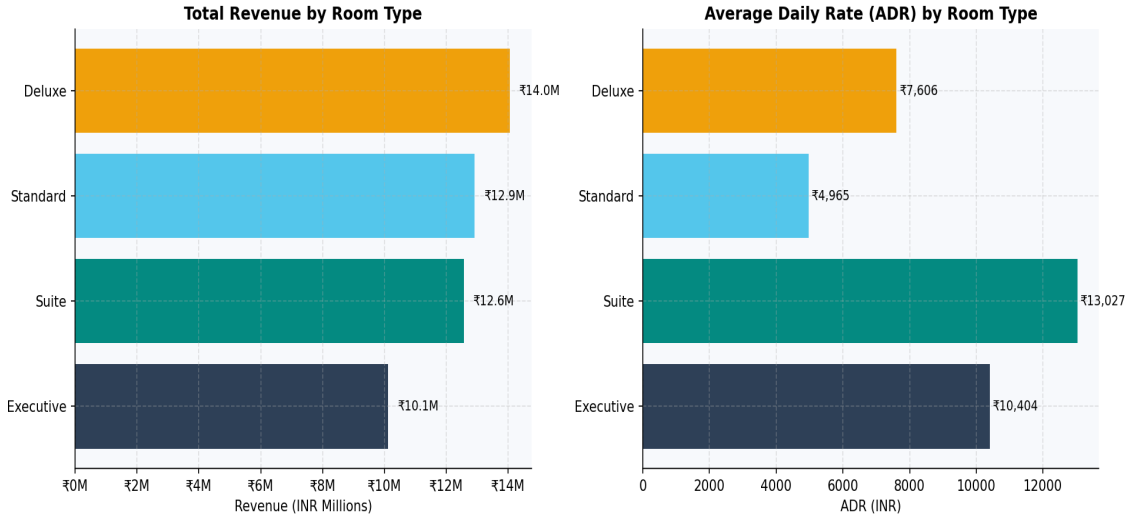

Fig 1.2 — Total Revenue and ADR by Room Type

Insight: Suites generate ~2.7x the ADR of Standard rooms. A targeted upsell strategy at check-in could significantly lift RevPAR.

2. RevPAR & Occupancy Insights

RevPAR (Revenue Per Available Room) is the gold standard hospitality KPI. It combines both ADR and occupancy, giving a holistic view of revenue performance. The heatmap further reveals which room types drive bookings in each month.

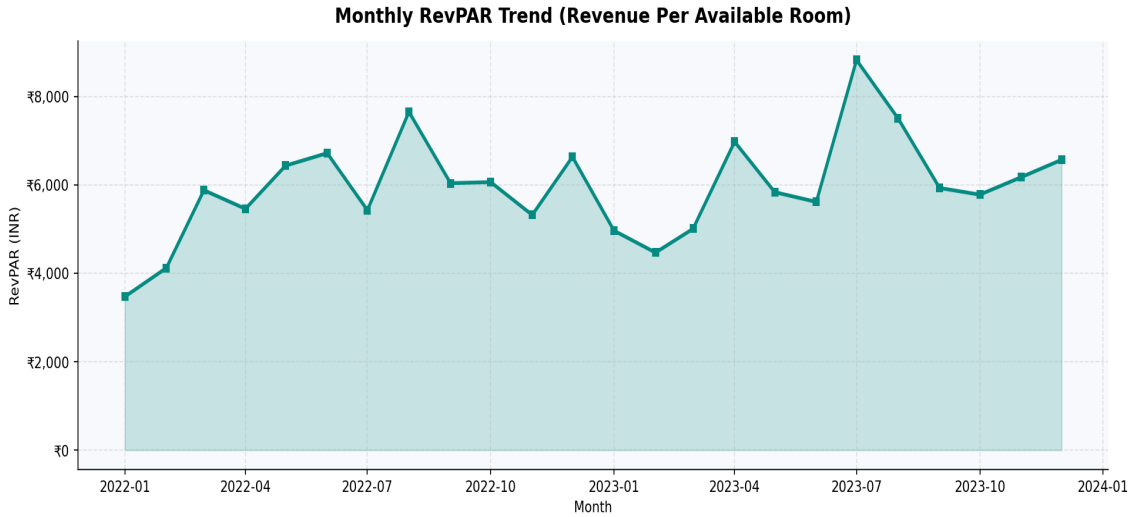

Fig 2.1 — Monthly RevPAR Trend

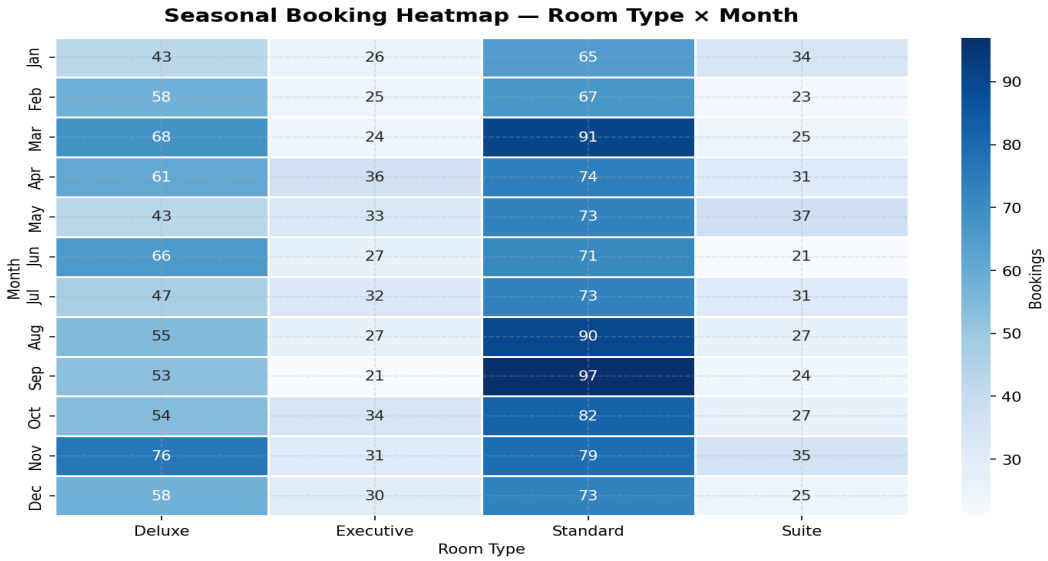

Fig 2.2 — Seasonal Booking Heatmap (Room Type × Month)

Insight: Standard rooms dominate bookings year-round. Suite and Executive bookings concentrate in Q3 (Jul–Sep), suggesting a premium leisure travel segment.

3. Cancellation Analysis

The overall cancellation rate stands at 23.2%. OTA (Online) bookings exhibit the highest cancellation rate, reflecting flexible cancellation policies offered by third-party platforms. Walk-in guests have negligible cancellations.

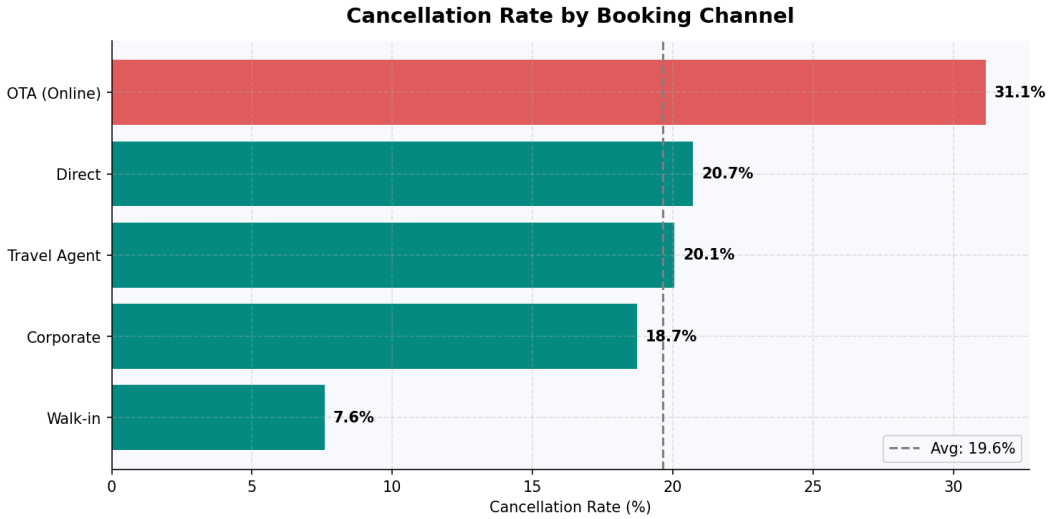

Fig 3.1 — Cancellation Rate by Booking Channel

Booking Channel	Cancellation Rate	Risk Level
Walk-in	7.6%	Low
Corporate	18.7%	Low
Travel Agent	20.1%	Medium
Direct	20.7%	Medium
OTA (Online)	31.1%	High

Insight: Shifting 10% of OTA bookings to Direct channel could reduce cancellations and save ~5–8% in commission fees, directly boosting net revenue.

4. Guest Satisfaction & Segmentation

Guest satisfaction ratings (1–10) were collected from non-cancelled bookings. Suite and Executive rooms lead satisfaction scores, while Direct channel bookings outperform OTA bookings, reinforcing the value of owning the guest relationship.

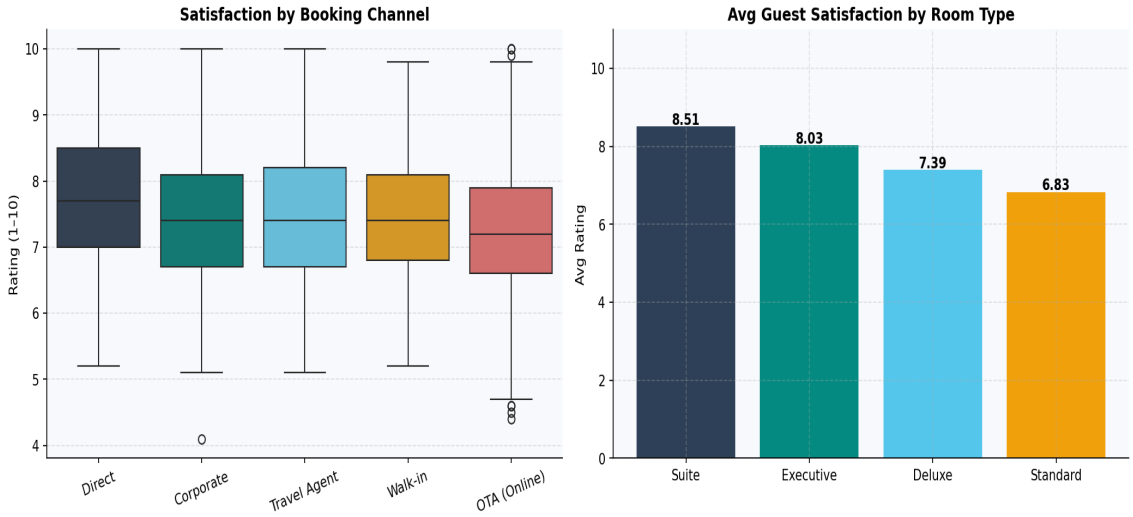

Fig 4.1 — Guest Satisfaction by Channel and Room Type

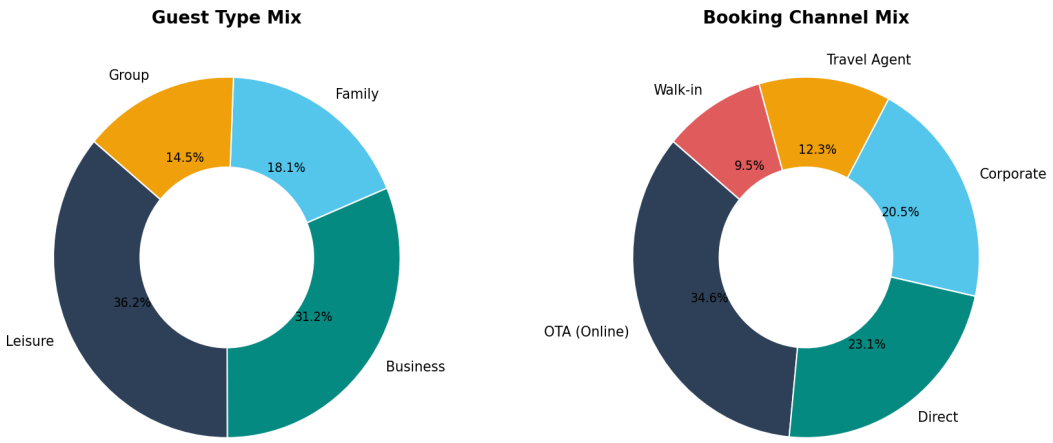

Fig 4.2 — Guest Type Mix and Booking Channel Mix

Insight: Repeat guests represent 27.1% of bookings. A structured loyalty programme targeting Business and Leisure guests could raise this to 35%+, reducing acquisition costs.

5. Cancellation Prediction Model

A Random Forest classifier was trained on 11 features to predict booking cancellations. The model enables revenue management teams to identify high-risk bookings early and deploy targeted retention strategies such as personalised communication or overbooking adjustments.

ROC-AUC Score	CV Mean AUC	Top Feature	Feature Importance
0.6195	**0.5766**	**Average Daily Rate**	**20.8%**

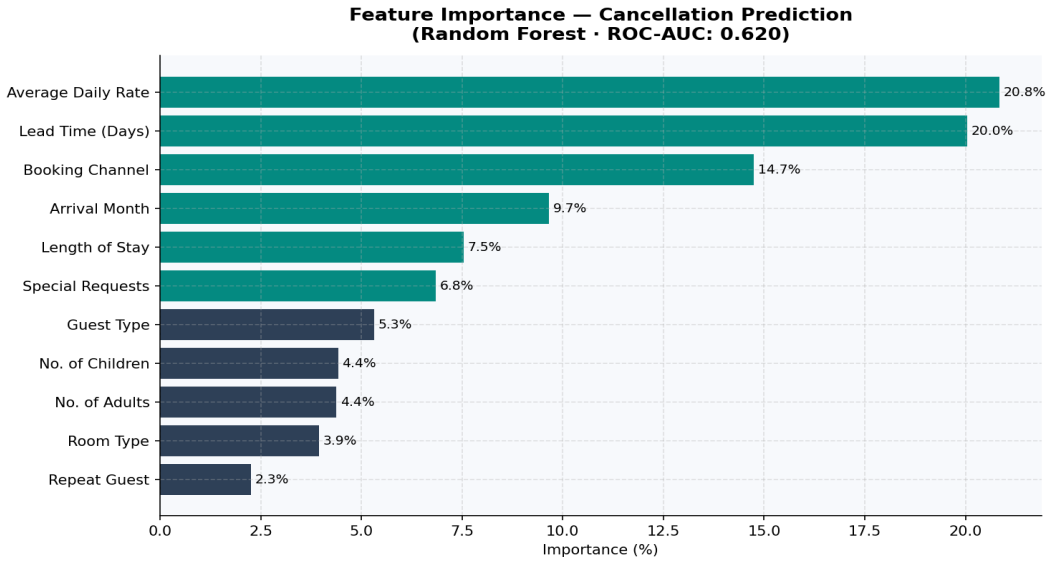

Fig 5.1 — Feature Importance (Random Forest)

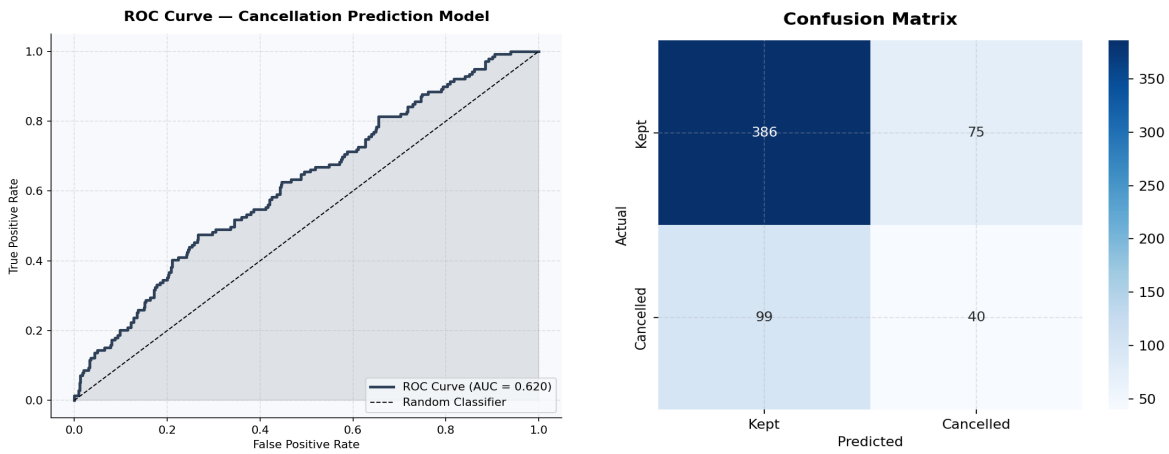

Fig 5.2 — ROC Curve (left) and Confusion Matrix (right)

Insight: Lead Time (Days) is the strongest cancellation predictor (20.8% importance). Bookings made >60 days in advance should trigger automated retention workflows.

6. Correlation & Lead Time Insights

The correlation matrix reveals key relationships between numerical variables. Total revenue correlates strongly with ADR and length of stay. Cancellations show the strongest positive correlation with lead time, validating the ML model's top feature finding.

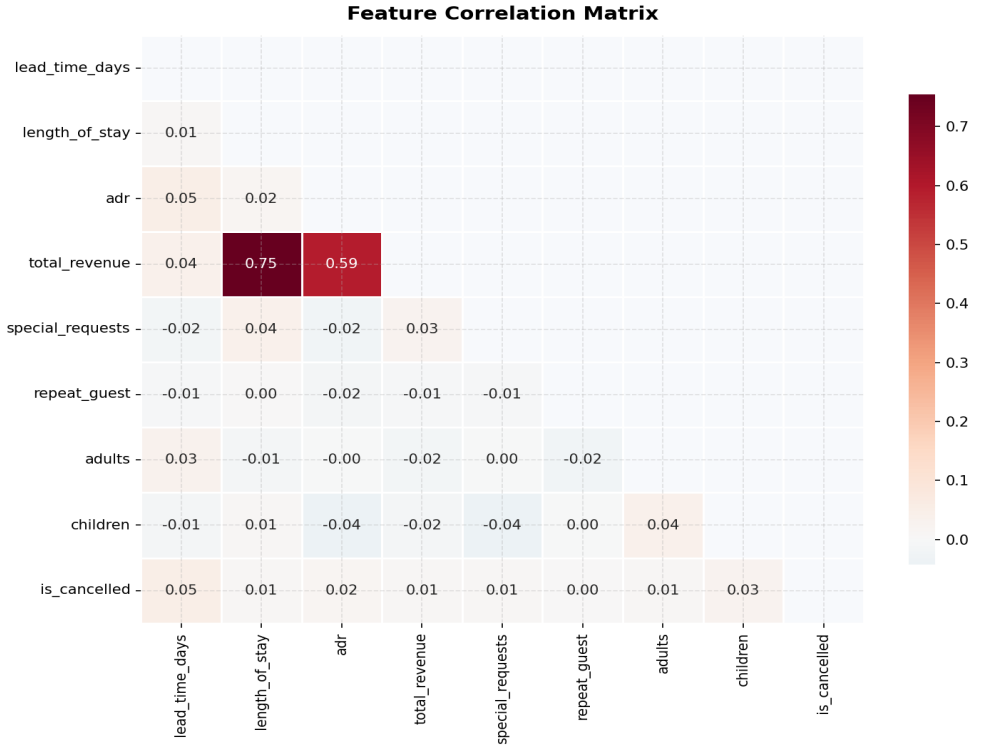

Fig 6.1 — Feature Correlation Matrix

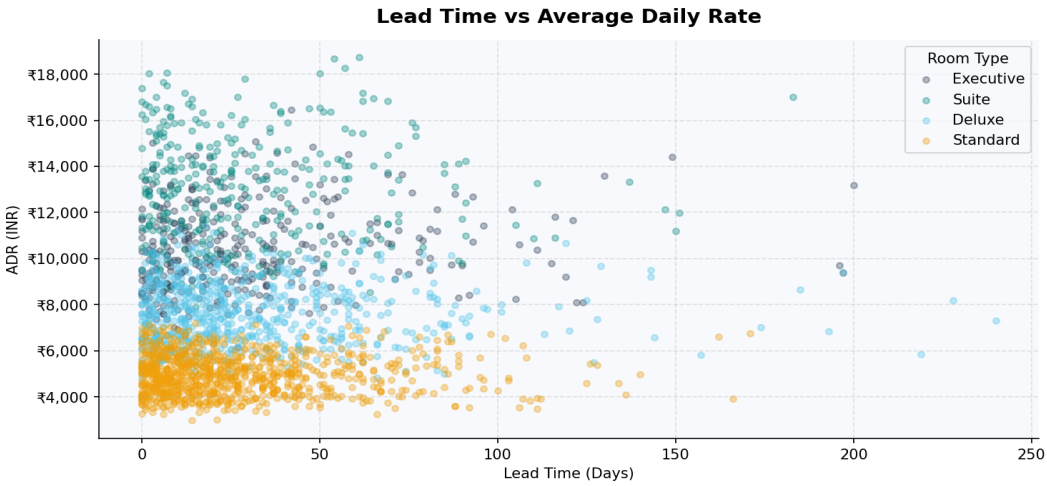

Fig 6.2 — Lead Time vs ADR by Room Type

Insight: No strong linear correlation between lead time and ADR — guests book early regardless of room tier. Yield-managed dynamic pricing could monetise early bookers more effectively.

7. Strategic Recommendations

Reduce OTA Dependency	Shift 10–15% of OTA volume to Direct channel via a Best Rate Guarantee and loyalty perks. This reduces ~30% cancellation risk and saves 15–20% OTA commission on migrated bookings.
Dynamic Pricing in Off-Peak Months	Deploy demand-based pricing in Jan–Feb and Oct–Nov. Targeted packages (e.g., weekend getaways, corporate retreats) can lift occupancy 8–12%.
Early Booking Retention Workflow	Flag bookings with lead time >60 days as high cancellation risk. Automated personalised emails (room upgrade teasers, local experience offers) can reduce cancellations by an estimated 20%.
Suite Upsell Programme	Train front-desk staff on upsell scripting. With an ADR differential of Rs 7,500+ vs Standard rooms, converting even 5% of Standard arrivals yields significant RevPAR uplift.
Loyalty & Repeat Guest Initiative	With 27.1% repeat guests currently, a tiered loyalty programme (Silver/Gold/Platinum) targeting Business and Leisure segments could raise repeat rate to 35%+ within 12 months.

8. Methodology & Tech Stack

Tool / Library	Version	Use Case
Python	3.11	Core programming language
Pandas	2.x	Data loading, cleaning, aggregation
NumPy	1.x	Numerical operations, synthetic data generation
Matplotlib	3.x	Custom chart rendering
Seaborn	0.x	Statistical visualisations (heatmap, boxplot)
Scikit-learn	1.x	Random Forest model, train-test split, metrics
ReportLab	4.x	Automated PDF report generation

Analytical Approach

- Data Generation — Synthetic dataset created with realistic seasonality, pricing curves, and behavioural patterns (3,000 records).
- Exploratory Data Analysis (EDA) — Distribution checks, segment breakdowns, and temporal trend analysis.
- Business KPI Calculation — ADR, RevPAR, cancellation rates, repeat guest ratios.
- Visualisation — 12 publication-quality charts covering revenue, occupancy, satisfaction, and ML outputs.
- Machine Learning — Binary classification (Random Forest) with cross-validation, feature importance, ROC-AUC evaluation.
- Report Generation — Automated PDF assembled programmatically using ReportLab.

This project was created using synthetic data for portfolio demonstration purposes. All insights and KPIs reflect modelled scenarios.